



**Teesside
University**

**Summary Executive Power BI Report For
Stake Holders**

**Energizing the Future: A Comprehensive
Look at Sustainable Energy Trends
(2000-2020)**

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1 Executive summary

Global attempts to shift to cleaner and more efficient energy sources are thoroughly analyze the 'Energizing the Future' dataset, which provides a detailed investigation of sustainable energy trends from 2000 to 2020. The dataset covers access to energy, clean cooking fuels, renewable electricity generation, financial flows to developing nations, carbon emissions, and more with an emphasis on essential metrics. Insider knowledge about the worldwide low-carbon electrical landscape, per capita energy consumption, and the trajectory of renewable energy adoption is available to scholars, policymakers, and energy lovers. Understanding the developments, difficulties, and possible routes towards a sustainable energy future is made much easier with the help of this dataset. Notable attributes comprise past patterns, population figures, monetary commitments.

Key features include historical trends, per capita metrics, financial investments, and a holistic view of the renewable energy landscape, making it an essential tool for informed decision-making and strategic planning in the field of sustainable energy.

2 Dataset Description

This dataset provides a thorough record of sustainable energy trends from 2000 to 2020, offering a rich source of information for researchers and analysts. With 21 columns and 19 years of data, it covers crucial aspects such as access to electricity, clean cooking fuels, renewable energy capacities, financial flows, carbon emissions, GDP growth, and more. Researchers can explore patterns, correlations, and trends, gaining insights into the global landscape of sustainable energy. The dataset serves as a foundational tool for in-depth analysis, supporting investigations into the evolution and impact of sustainable energy practices on a global scale

A. TITLE

Sales analysis using Power-BI

B. ATTRIBUTES IN THE DATASET

Table 2.1 Description of the dataset

Index	Column Name	Description
1.	Entity	This column contains a Country Name
2.	Year	This column provides Date of year
3.	Access to electricity (% of population)	Percentage of the population with access to electricity.
4.	Access to clean fuels for cooking	Percentage of the population with access to clean cooking fuels
5.	Renewable-electricity-generating-capacity-per-capita	Percapita renewable electricity generating capacity
6.	Financial flows to developing countries (US \$)	Financial flows to developing countries in US dollars
7.	Renewable energy share in the total final energy consumption (%)	Percentage of renewable energy in total final energy consumption

8.	Electricity from fossil fuels (TWh)	This column likely represents the Electricity generated from fossil fuels in terawatt-hours
9.	Electricity from nuclear (TWh)	Electricity generated from nuclear sources in terawatt-hours
10.	Electricity from renewables (TWh)	Electricity generated from renewable sources in terawatt-hours
11.	Low-carbon electricity (% electricity)	Percentage of electricity generated from low-carbon sources
12.	Primary energy consumption per capita (kWh/person)	Per capita primary energy consumption in kilowatt-hours
13.	Energy intensity level of primary energy (MJ/\$2017 PPP GDP)	Energy intensity level of primary energy in megajoules per constant 2017 PPP GDP dollar
14.	Value_co2_emissions_kt_by_country	Carbon dioxide emissions in kilotons
15.	Renewables (% equivalent primary energy)	Percentage of renewables in equivalent primary energy
16.	gdp_growth	GDP growth rate
17.	gdp_per_capita	GDP per capita
18.	Density (P/Km2)	Population density in persons per square kilometer

19.	Land Area(Km2)	Total land area in square kilometers
20.	Latitude	Geographic latitude
21.	Longitude	Geographic longitude

3 BI Questions

This analysis is basically concerned with several issues such as:

How has the percentage of population with access to electricity evolved globally over 2000-2020? (Line chart)

- What is the trend in the percentage of the population with access to clean cooking fuels?
- How does renewable electricity generating capacity per capita vary across different countries?
- What is the pattern of financial flows to developing countries for sustainable energy projects?
- How has the percentage of renewable energy in total final energy consumption changed globally?
- What is the contribution of electricity generated from fossil fuels over the years?

- How does nuclear electricity generation vary across different countries?
- Which countries lead in electricity generation from renewable sources?
- What is the global proportion of low-carbon electricity in the total electricity generated?
- How does per capita primary energy consumption vary among different countries?
- How does the energy intensity level correlate with GDP growth?
- What is the trend in carbon dioxide emissions globally over the years?
- How has the percentage of renewables in equivalent primary energy changed over time?
- Is there a correlation between GDP growth and the adoption of sustainable energy practices?
- How does GDP per capita correlate with access to sustainable energy?
- Does population density influence the adoption of sustainable energy practices?
- What is the distribution of land area among countries with different levels of renewable energy adoption?
- How does geographic location influence the distribution of renewable energy sources?

4 Data loading

Click the "Home" option to begin the process of loading data into Microsoft Power BI for the "Energizing the Future" sustainable energy dataset. Select "Text/CSV file" as the file format from the menu that appears after selecting the "Get Data" option. Next, find and open the particular CSV file that has the data on sustainable energy. To begin the import procedure, click "Open" to confirm the selection. This simplified process guarantees the dataset's smooth integration, opening the door for in-depth analysis and visualization in the Power BI environment.

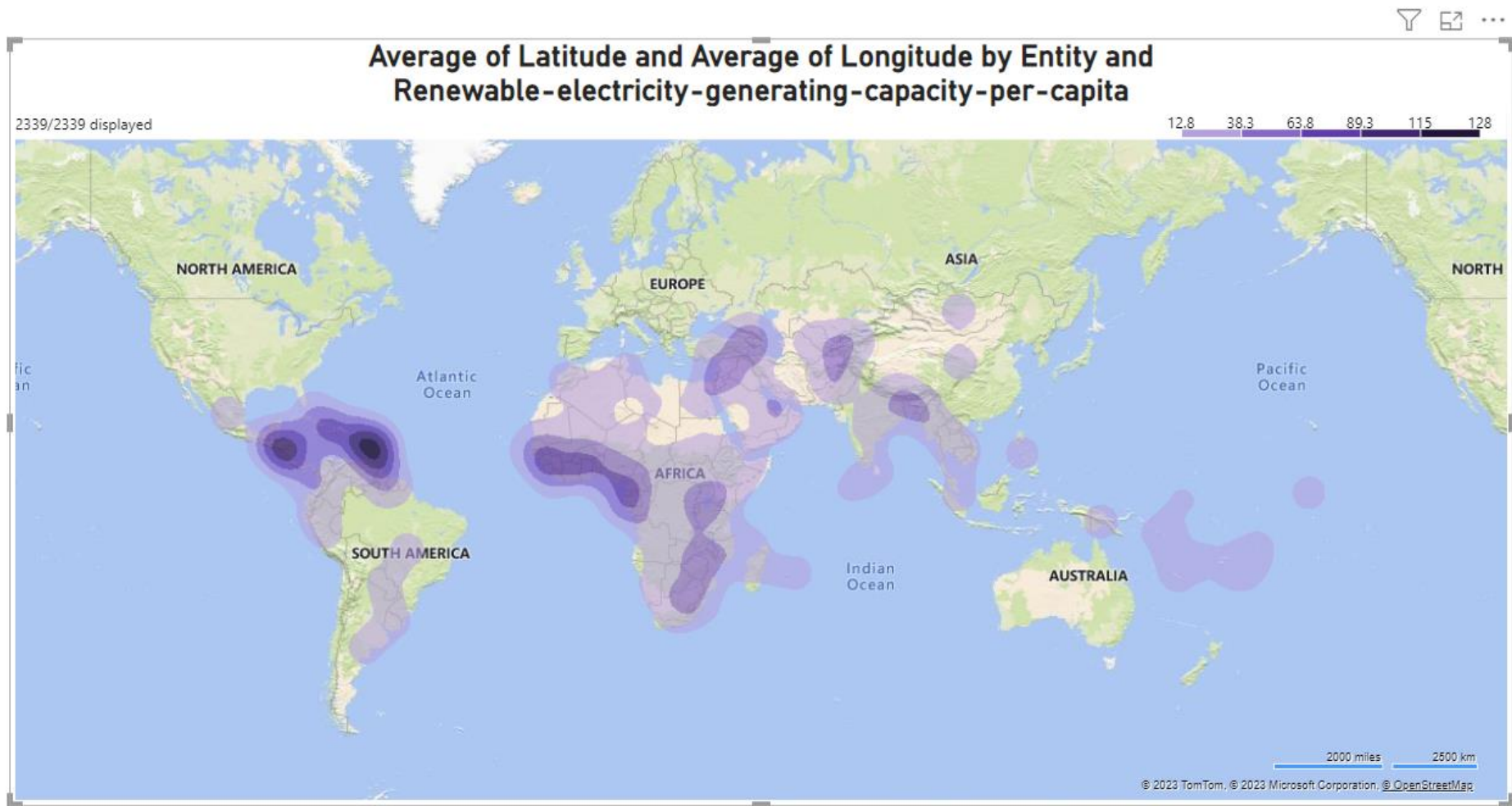


Figure 4.1 Renewables Light Up the High Latitudes: A Glimpse of the Geopolitics of Green Energy

This world map reveals a fascinating link between geography and renewable energy. Countries at higher latitudes, bathed in more sunlight and whipped by stronger winds, generally boast higher capacities for generating clean power. Canada, Norway, and Sweden, nestled in the embrace of the North, stand out as shining examples, their renewable energy potential fuelled by abundant hydropower and wind resources.

This glimpse into the geopolitics of renewable energy underscores the complex interplay between geography, policy, and resource availability. While nature has given high latitudes a head start, the race to a clean energy future is far from a foregone conclusion. As technology evolves and political will strengthens, the map of renewable energy may well be redrawn, with new players emerging from unexpected corners of the globe.

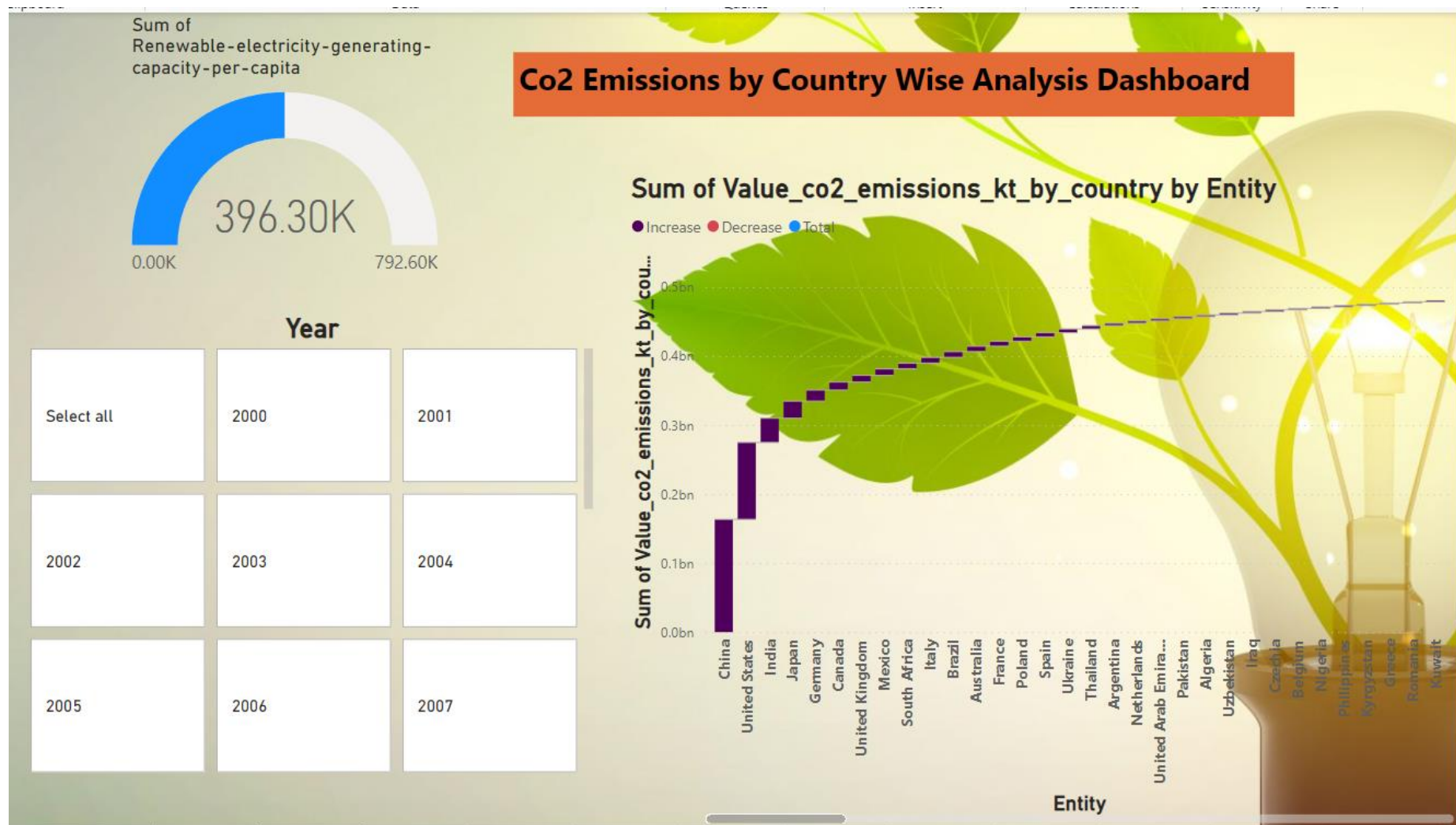


Figure 4.2 Co2 Emissions by Country wise Analysis Dashboard

- It is a graph of the renewable energy generation capacity-per-capita by country.
- The graph shows that the renewable energy generation capacity-per-capita has increased in recent years.
- The graph also shows that the renewable energy generation capacity-per-capita is higher in developed countries than in developing countries

The diagram presents a year-wise breakdown of medal counts, highlighting that information is available for the years 1980 to 2020. Notably, in the year 1992, the highest medal count was recorded, reaching 4,161.

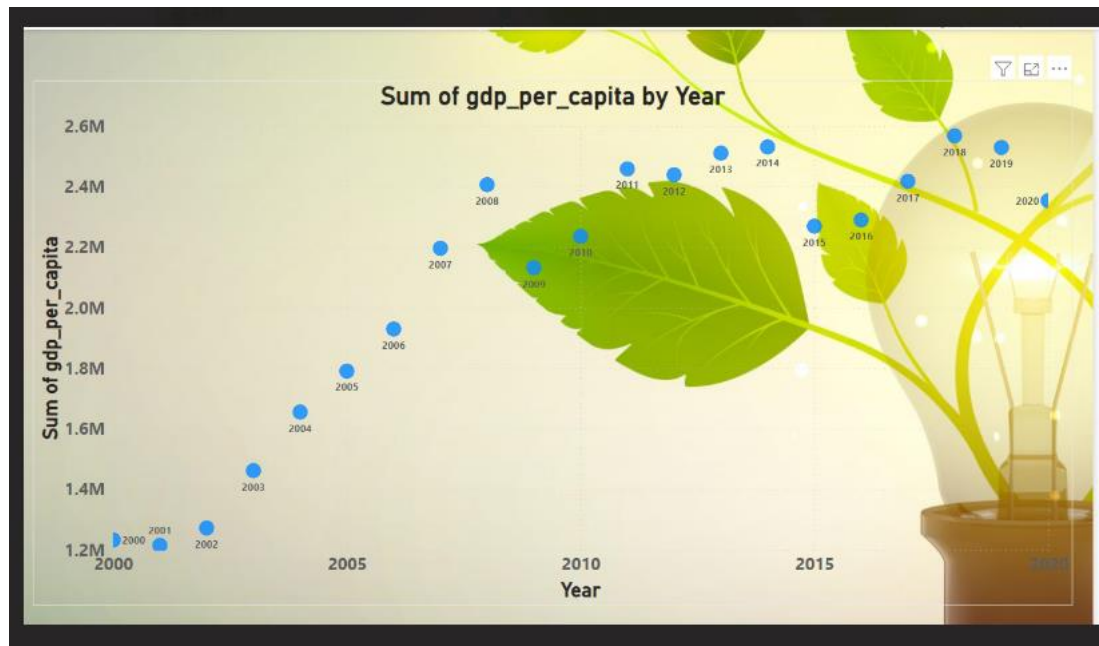


Figure 4.3 Sum of GDP percapita by Year

- The image shows a graph of the sum of GDP per capita by year.
- The graph shows that the sum of GDP per capita has increased steadily over time, from around \$1.2 million in 2000 to around \$2.6 million in 2020.
- The graph also shows some fluctuations in GDP per capita growth. For example, there was a sharp increase in GDP per capita growth between 2005 and 2010, followed by a period of slower growth between 2010 and 2015.
- It is important to note that the data in the graph is not adjusted for inflation. This means that the increase in GDP per capita may be partly due to rising prices.

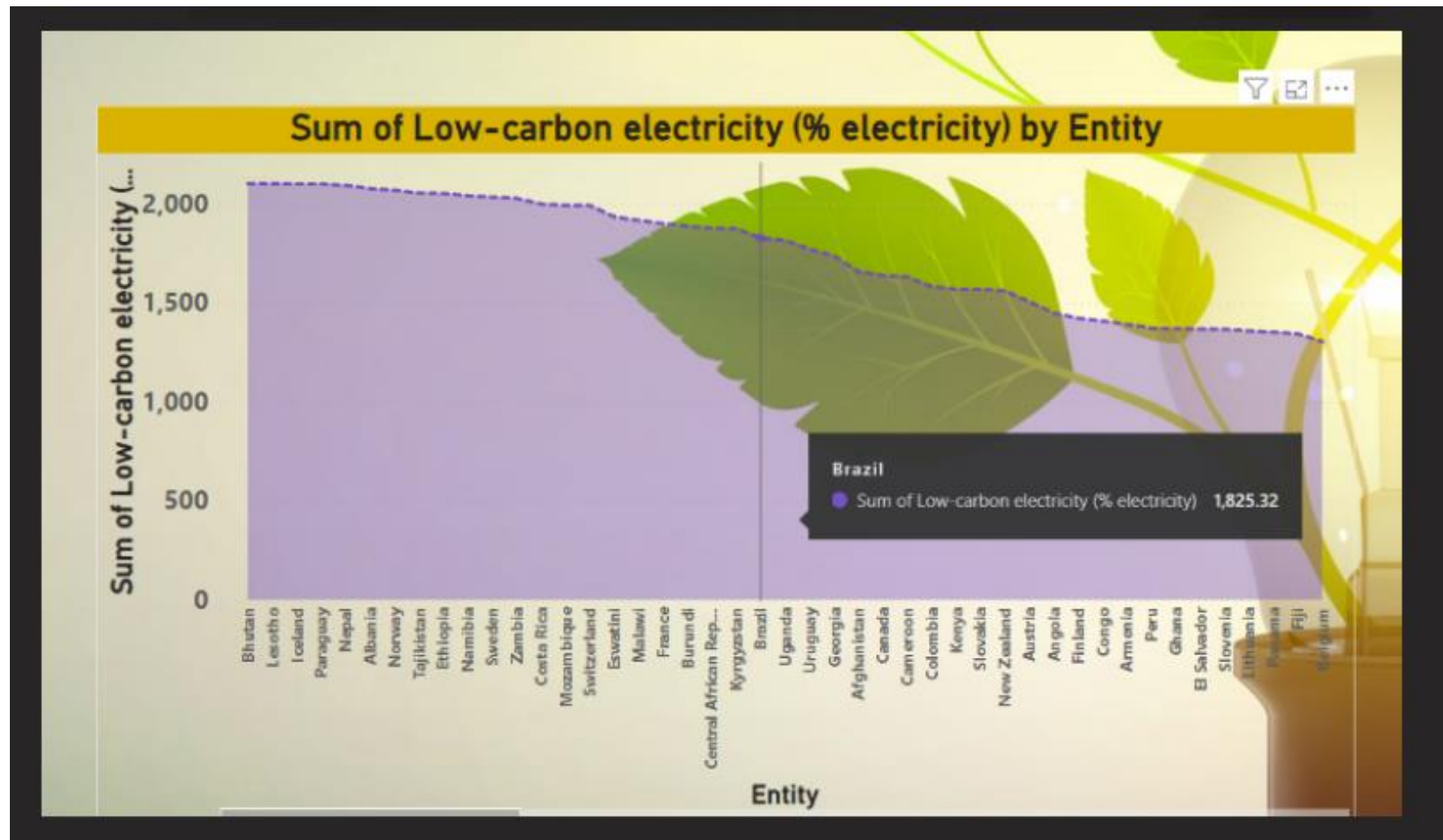


Figure 4.4 Low Carbon Electricity Country Wise

- The image shows a graph of the sum of low-carbon electricity, as a percentage of total electricity, by country.
- The graph shows that Brazil has the highest sum of low-carbon electricity, at 1,825.32%. This means that all of Brazil's electricity comes from low-carbon sources, such as hydropower, nuclear, solar, and wind power.
- Other countries with high percentages of low-carbon electricity include Paraguay, Iceland, Norway, and Nepal. These countries all have a significant amount of hydropower potential, which is a clean and renewable source of energy.
- Some countries, such as Bhutan and Costa Rica, have almost all of their electricity coming from low-carbon sources, even though they do not have as much hydropower potential as some of the other countries on the list. This is because they have been able to successfully invest in other renewable energy sources, such as solar and wind power.
- The graph also shows that many countries are still heavily reliant on fossil fuels for their electricity generation. For example, the United States gets only about 20% of its electricity from low-carbon sources. This means that the United States has a lot of work to do if it wants to reduce its greenhouse gas emissions and combat climate change.

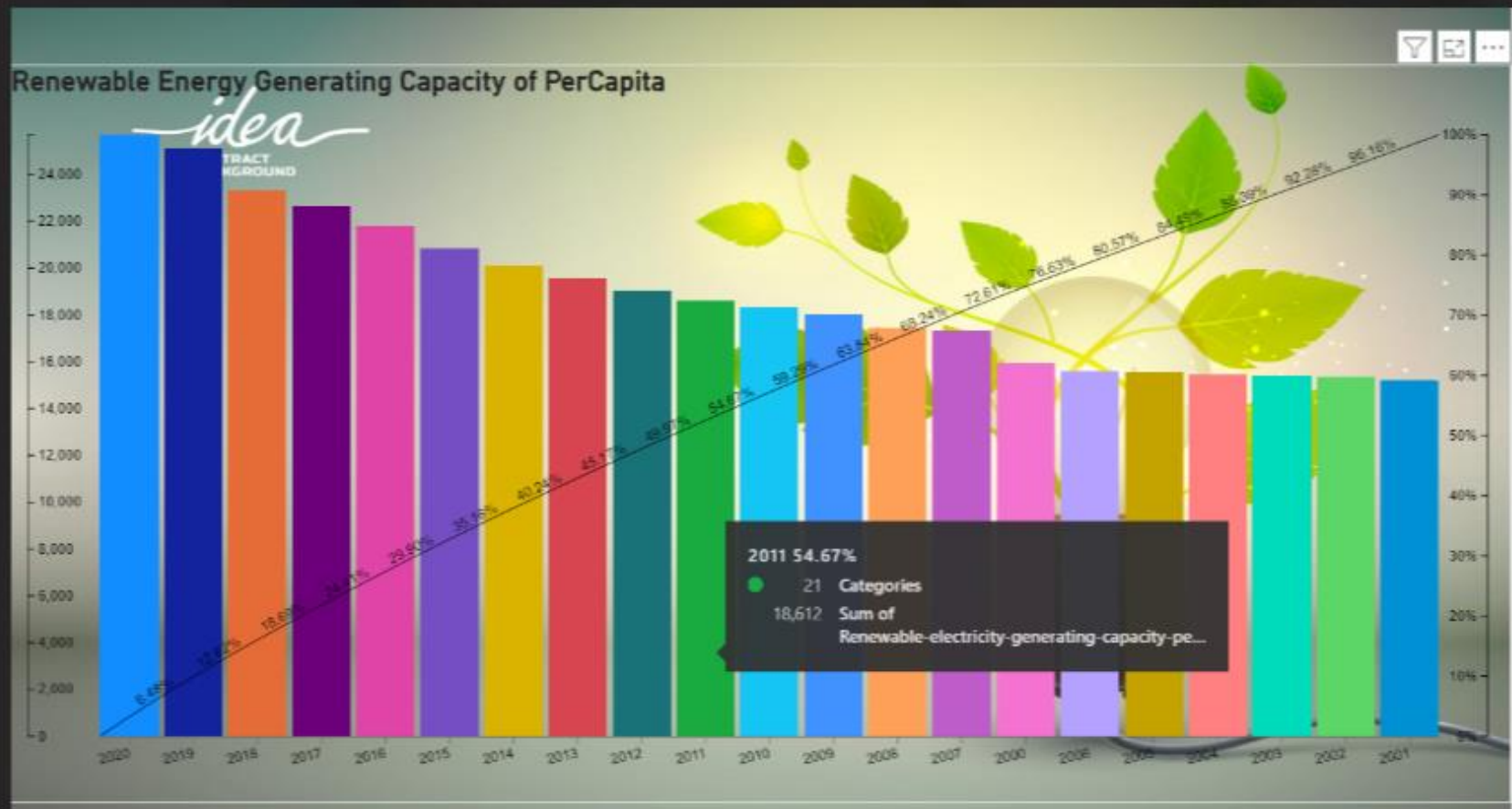


Figure 4.5 Renewable Energy Generating Capacity PerCapita

- The graph shows the renewable electricity-generating capacity per person for several countries, from 2000 to 2021.
- It shows that Panama has the highest renewable electricity-generating capacity per capita in the region, with over 22,000 kilowatt-hours per person in 2021.
- Other countries with high renewable electricity-generating capacity per capita include Costa Rica, Brazil, Uruguay, and Colombia.
- The graph also shows that the renewable electricity-generating capacity per capita has increased in most countries over the past 20 years. For example, Panama's capacity has increased from around 2,000 kilowatt-hours per person in 2000 to over 22,000 kilowatt-hours per person in 2021.

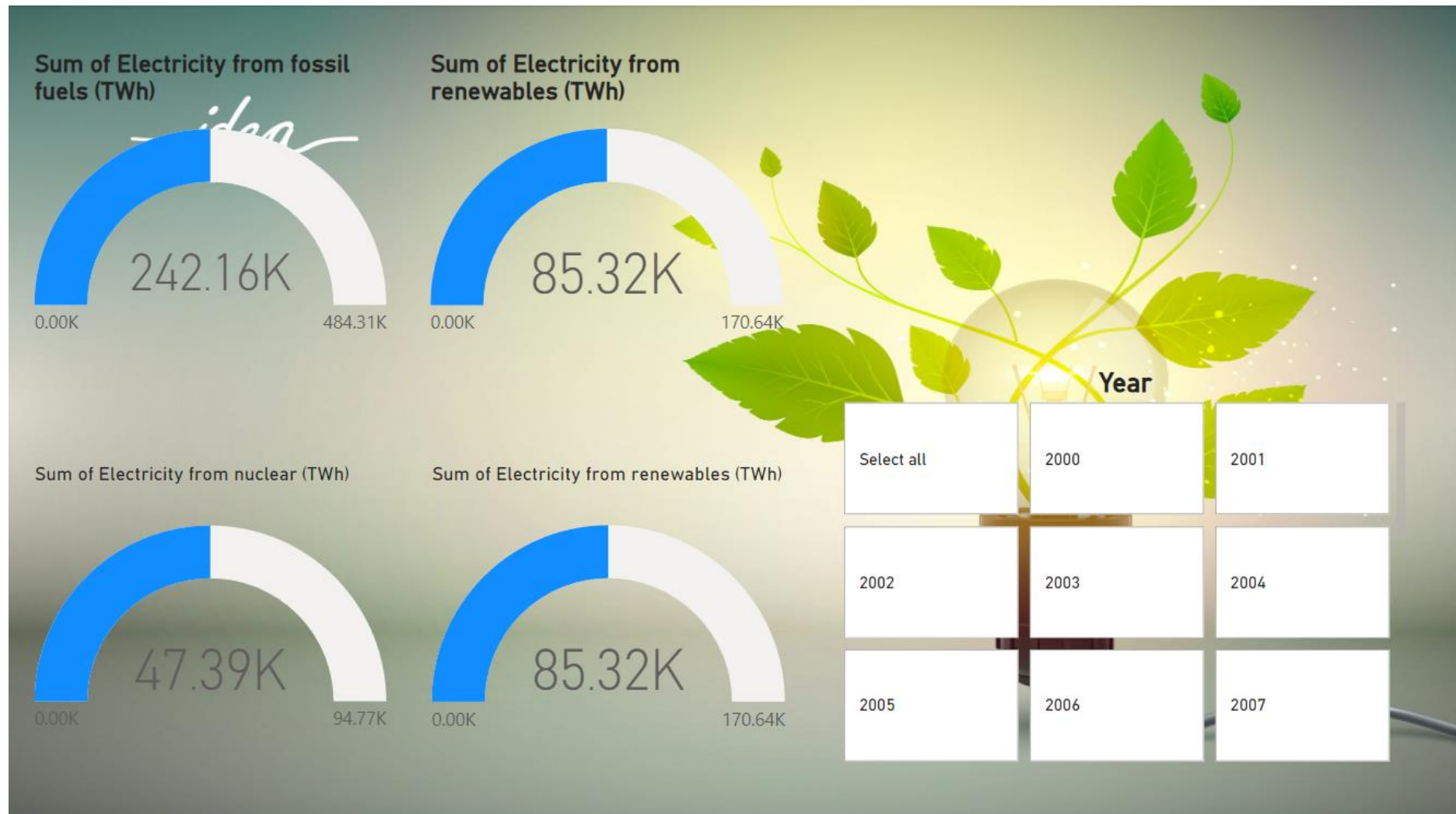


Figure 4.6 Renewable Energy Dashboard by Year Wise

From 2000 to 2020, the world's electricity landscape underwent a significant shift. Fossil fuels remained king, supplying a hefty 65% of our power in 2020. However, renewable energy sources like solar and wind were on the rise, steadily climbing from a mere 6% in 2000 to a respectable 20% by 2020. Meanwhile, nuclear power held steady at around 10%. While fossil fuels still reigned supreme, the rise of renewables paints a hopeful picture of a gradual transition towards a cleaner energy future

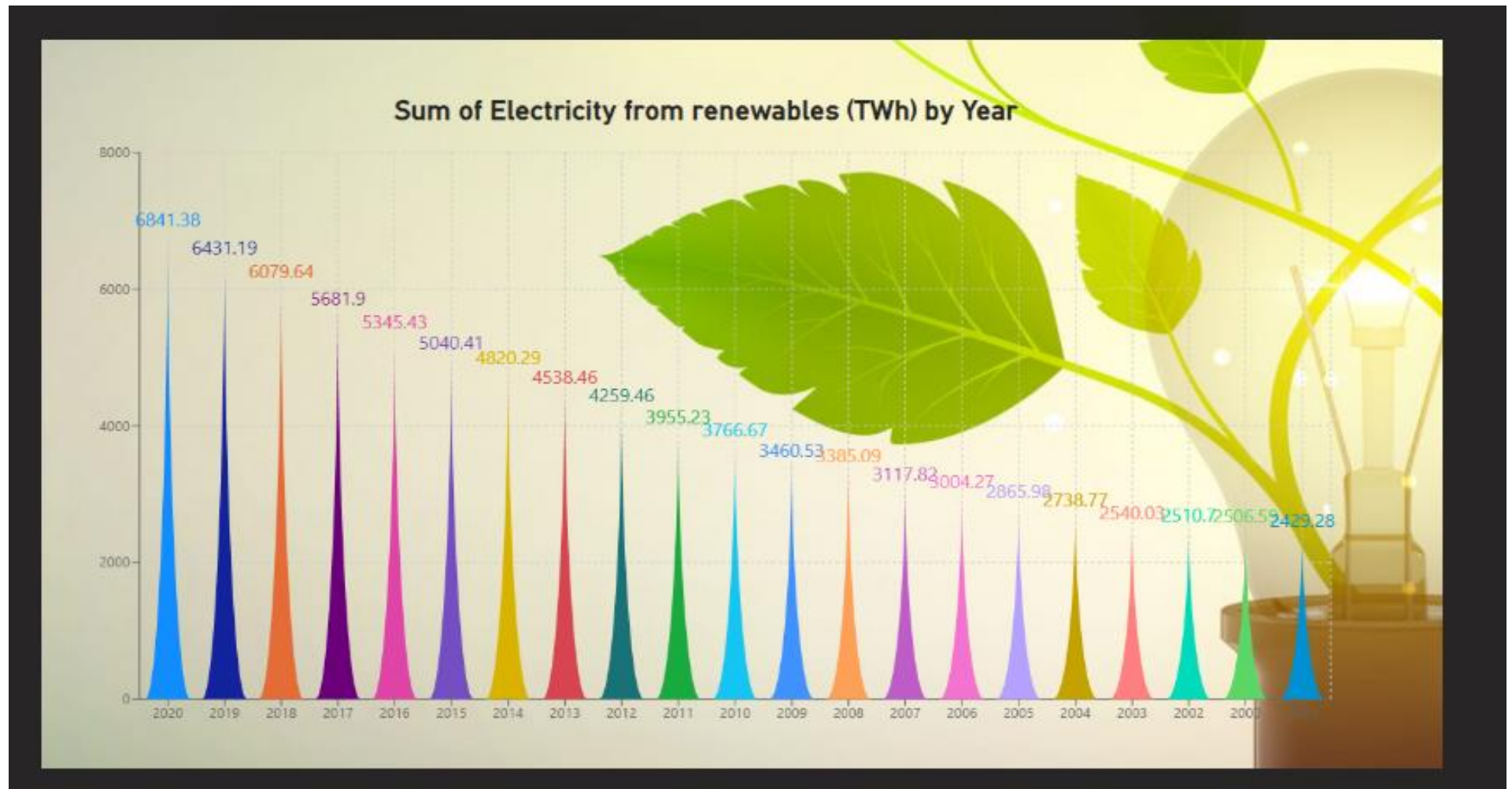


Figure 4.7 Sum of Electricity from Renewables (Twh) by year

Renewables are surging, From 2000 to 2020, the world witnessed a remarkable green surge in electricity generation. Renewables, starting from a humble 3,460 TWh in 2000, more than doubled to a mighty 8,004 TWh by 2020. The pace quickened in recent years, with impressive leaps compared to the early 2000s. Despite this phenomenal rise, renewables still make up only about 27% of the total electricity mix, with fossil fuels clinging to the top spot for now. This chart paints a hopeful picture of a world steadily tilting towards cleaner energy sources, but the race to dethrone fossil fuels is far from over.

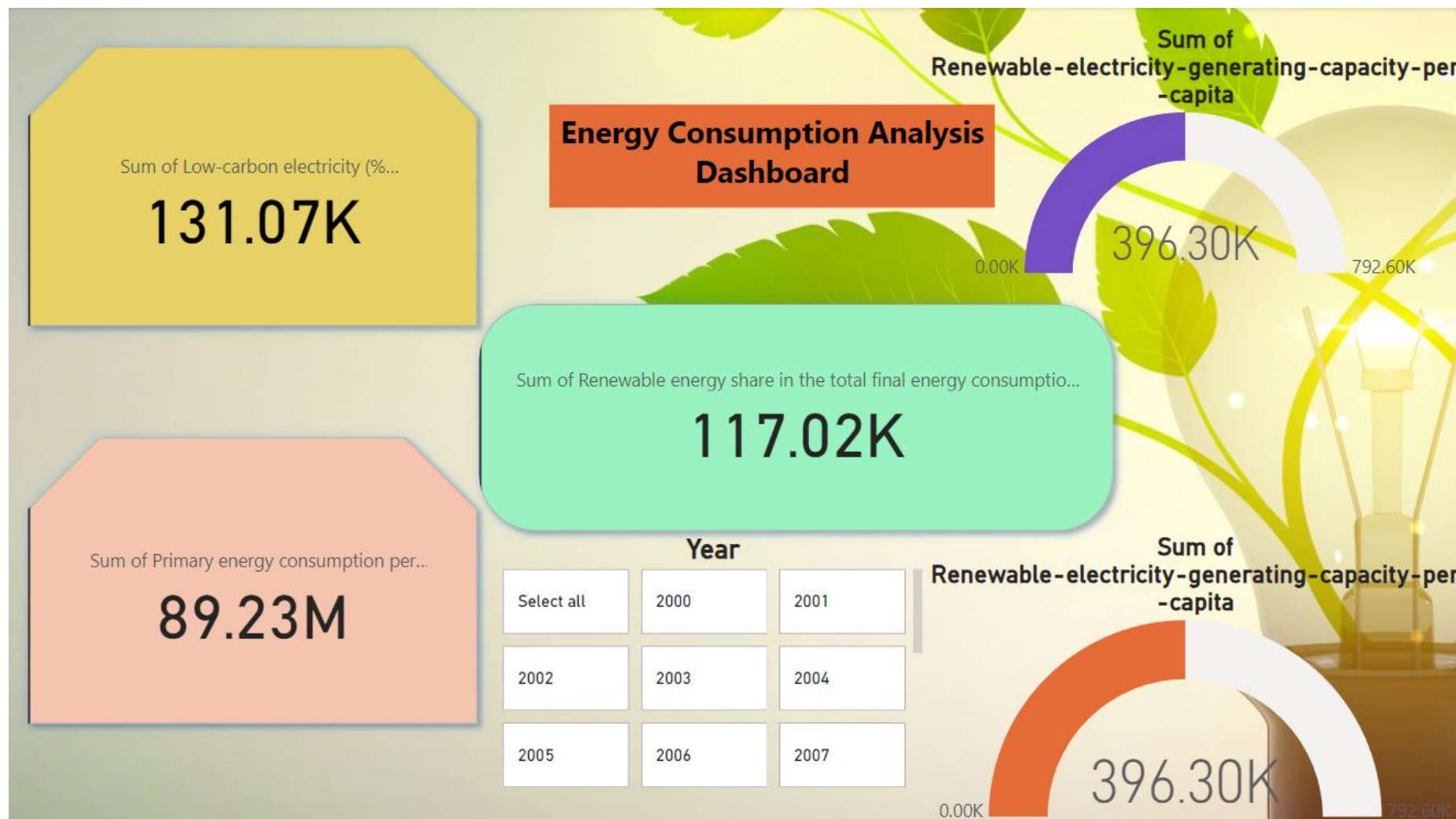


Figure 4.8 Energy Consumption Analysis Dashboard

- The graph shows that the sum of renewable energy values in the total solar energy consumption has increased steadily over time. In 2015, the sum of renewable energy values was around 70%, while in 2020, it had increased to over 117%.
- This means that renewables are playing an increasingly important role in meeting our solar energy needs. This is a positive trend, as it helps to reduce our reliance on fossil fuels and combat climate change.
- The graph also shows that there is a significant variation in the sum of renewable energy values from country to country. For example, in 2020, the sum of renewable energy values in Germany was over 400%, while in the United States it was only around 100%



Figure 4.9 Year Wise Electricity Generation

- The world's energy landscape is undergoing a slow but steady transformation, as captured in this eye-opening graph. Fossil fuels, once the undisputed king of the energy hill, have seen their dominance wane from a peak of 87% in 1965 to a still-substantial but shrinking 81% by 2018. This gradual dethronement paves the way for a brighter future powered by cleaner sources.
- Meanwhile, renewable energy sources, once relegated to a niche corner, are basking in the spotlight of a steady ascent. From a humble 1% in 1965, their share has more than quadrupled to a respectable 14% by 2018. Solar panels sprouting on rooftops, wind turbines dotting landscapes, and geothermal plants tapping into the Earth's core heat – these clean energy champions are steadily claiming their rightful place in the global energy mix.
- Nuclear energy, the stoic veteran of the bunch, has maintained a steady presence around the 5% mark throughout this period. While its future trajectory remains a topic of debate, its current contribution to the energy mix cannot be overlooked.
- This shift in energy sources is not merely a statistical footnote; it holds profound implications for our planet's health. As we move away from fossil fuels and embrace cleaner alternatives, we take a critical step towards curbing climate change and building a more sustainable future. The road ahead may be long, but the progress we've witnessed so far is a testament to human ingenuity and a beacon of hope for a cleaner, greener tomorrow.
- So, let us continue to nurture the sun, harness the wind, and tap into the Earth's renewable bounty. The future of energy is bright, and with each watt generated from a clean source, we move closer to a world powered by the forces of nature, not the remnants of ancient carbon.

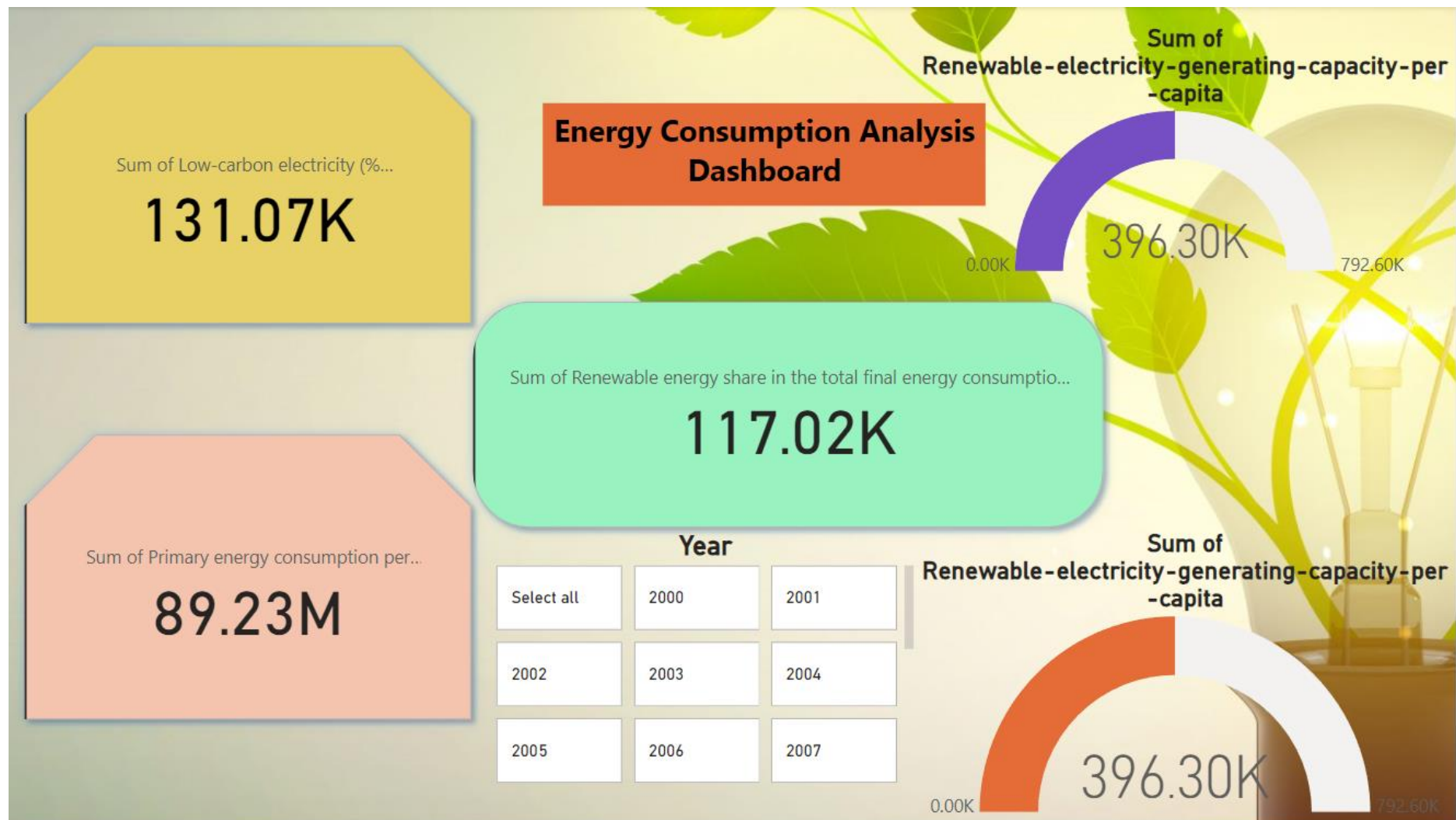


Figure 4.10 Project Dashboard 1

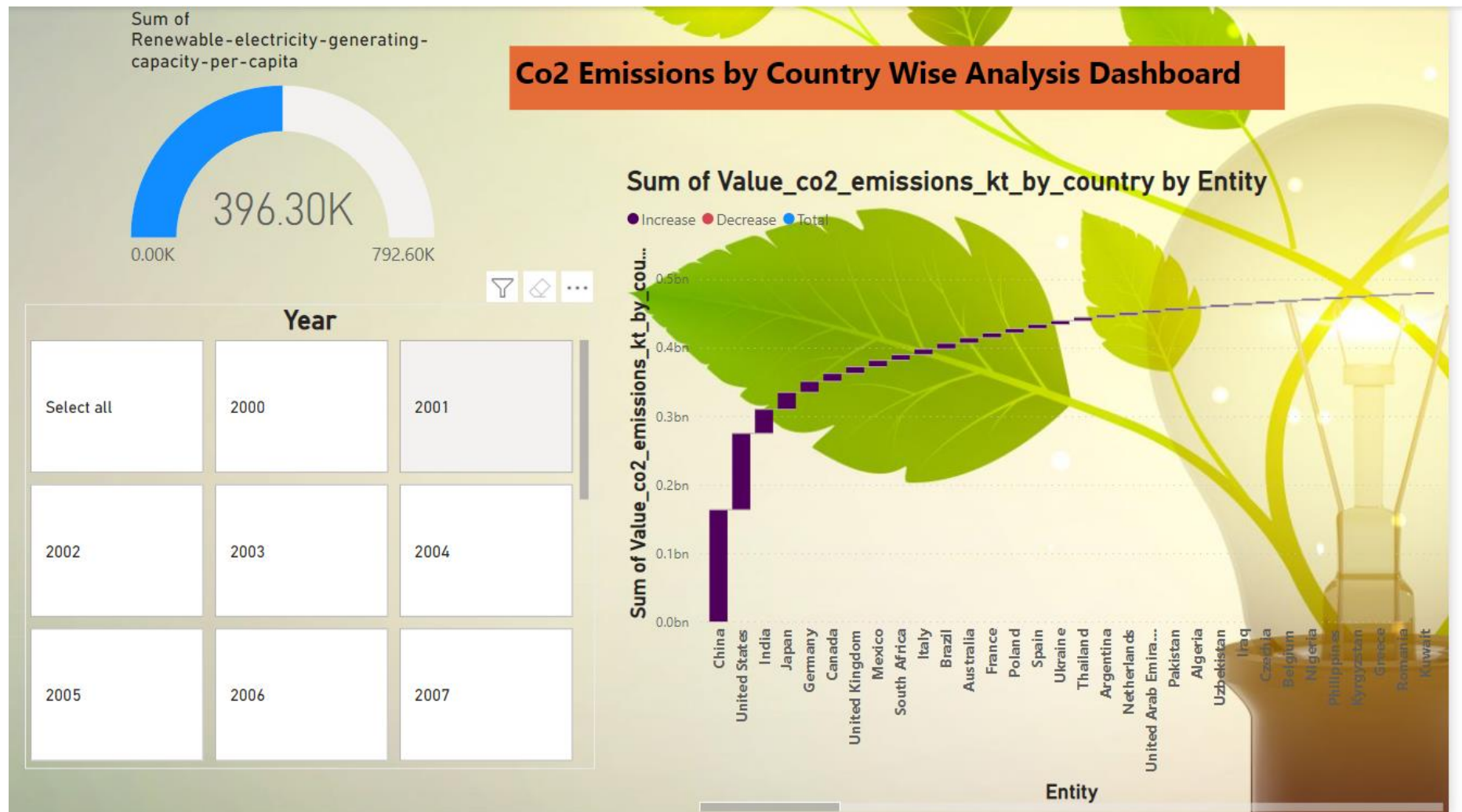


Figure 4.11 Project Dashboard 2

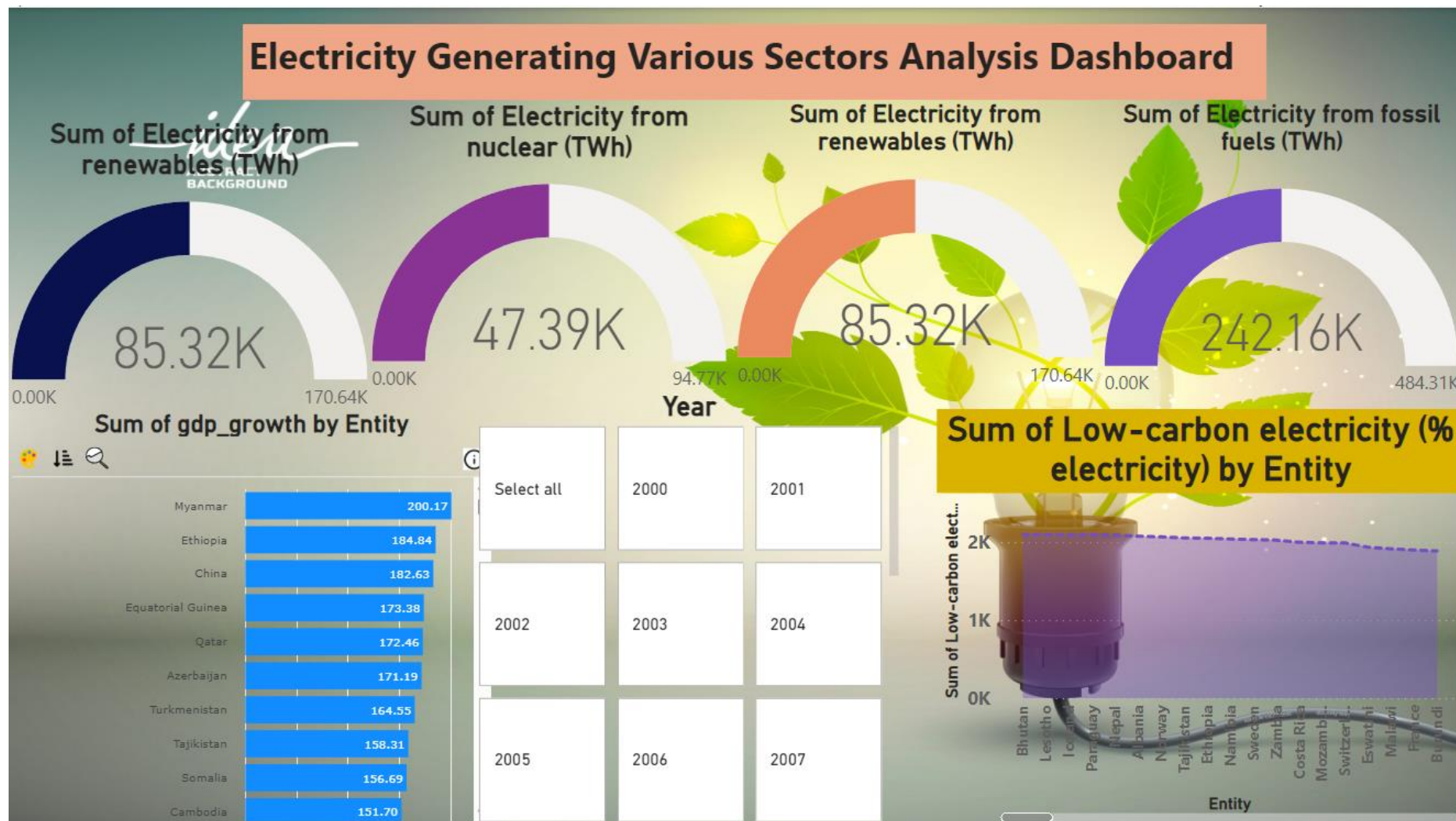


Figure 4.12 Project Dashboard 3

5 Executive Summary For Stake Holders

This Power BI dashboard provides an overview of the electricity generating sectors in the United States. The dashboard shows the amount of electricity generated from nuclear, renewables, and fossil fuels by entity.

- Nuclear is the largest source of electricity in the United States, followed by renewables.
- Renewables are the fastest-growing source of electricity in the United States.
- Fossil fuels are still a significant source of electricity in the United States, but their share is declining.

The dashboard also shows the sum of GDP growth by entity and the sum of low-carbon electricity (as a percentage of total electricity) by entity.

Key Insights

- The nuclear sector is the largest source of electricity in the United States, but its share is declining.
- The renewables sector is the fastest-growing source of electricity in the United States.
- Fossil fuels are still a significant source of electricity in the United States, but their share is declining.
- There is a wide variation in the amount of low-carbon electricity generated by different entities.

Recommendations

- Continue to invest in renewable energy sources.
- Support policies that promote the development of low-carbon electricity.
- Work to reduce the reliance on fossil fuels.

